

Quantum Toom's Contours

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Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math

Our Code

Code

Consider the representation of the d -dimensional toric by the cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ ¹.

In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

¹where 1_i is the vector consisting of one at the i th coordinate and zero elsewhere

Local Decoder

We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1 \neq g_2 \in S$, if $x = v$ and y is either $g_1 v$ or $g_2 v$.

definition

We say that D is a local decoder if when applied against $2p(1 - p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1 - p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.